INCH-POUND

MIL-C-83723/90C

30 June 1992

SUPERSEDING
MIL-C-83723/90B
6 July 1983

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SOLDER FLANGE MOUNT, THREADED COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H, N, AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

(C) The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83723.

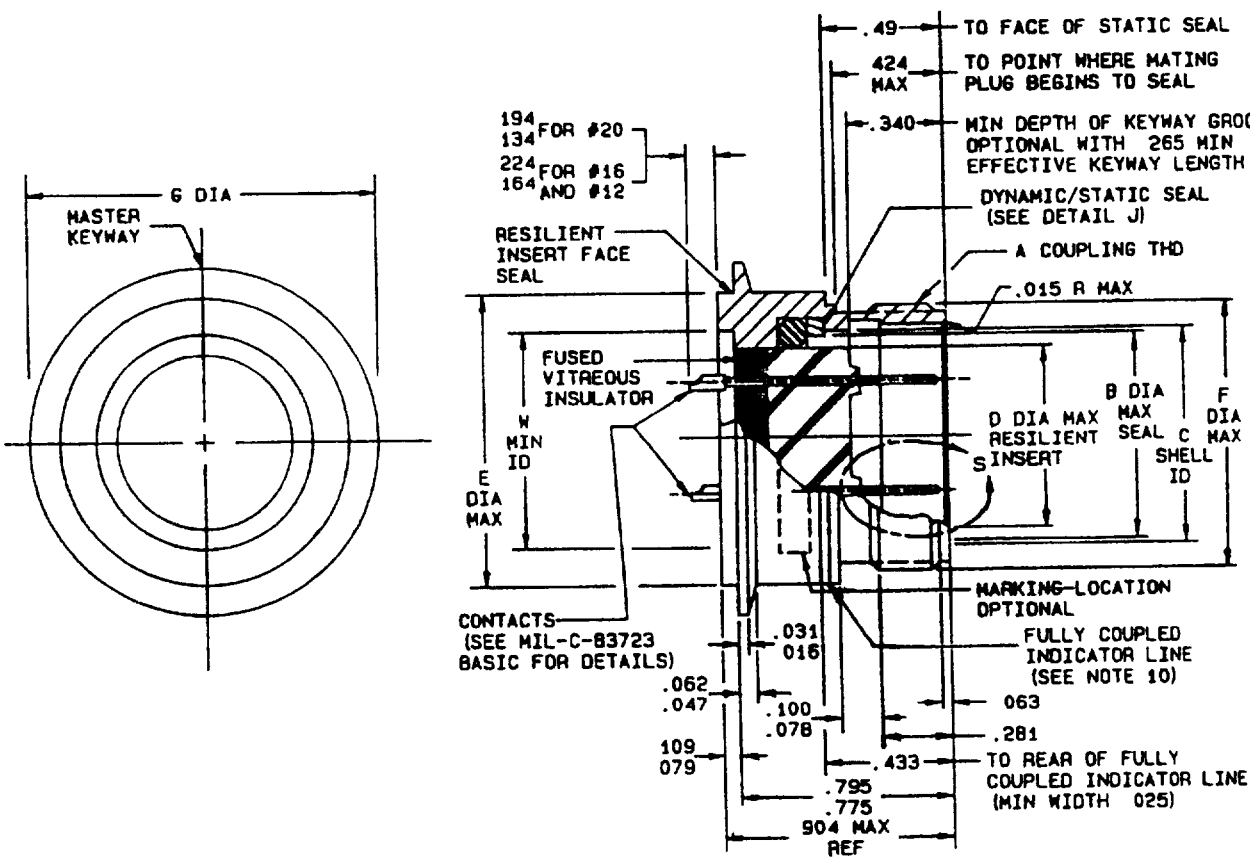


FIGURE 1. Dimensions and configurations.

(C) denotes changes

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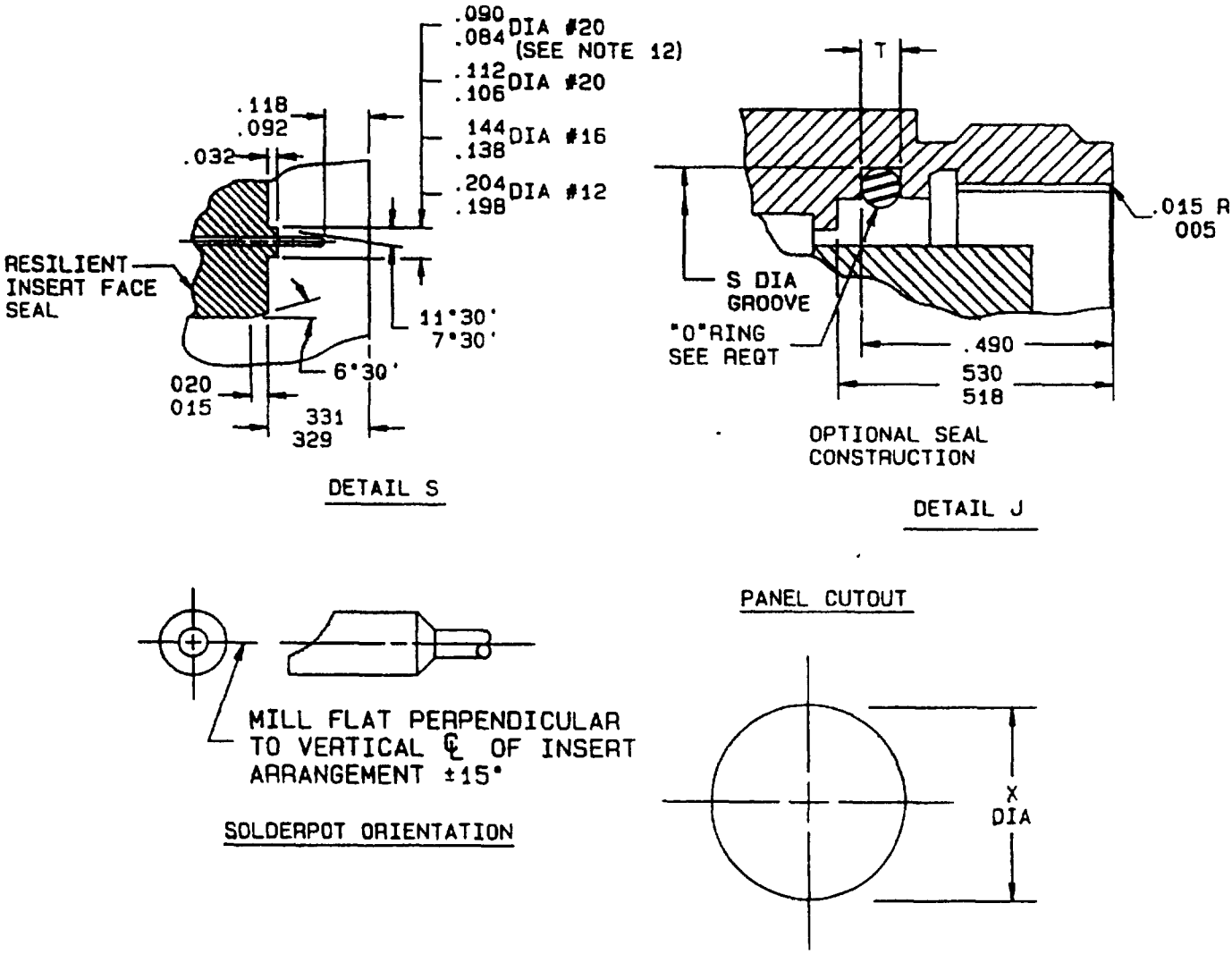


FIGURE 1. Dimensions and configurations - Continued.

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Connectors dimensions.

Shell size	(C) A Thread UNEF-2A	B	C	D	E	F
8	.562-24	.394	.418 .413	.290	.500 .494	.562
10	.688-24	.515	.535 .530	.388	.562 .556	.696
12	.875-20	.685	.700 .705	.558	.750 .744	.875
14	.938-20	.754	.774 .769	.627	.812 .806	.936
16	1.062-18	.881	.901 .896	.772	.937 .931	1.062
18	1.188-18	.987	1.007 1.002	.860	1.062 1.056	1.187
22	1.438-18	1.237	1.257 1.252	1.110	1.312 1.306	1.437

Connector dimensions - Continued.

Shell size	G	W	X	S +.005 -.000	T
8	.723 .703	.400	.510	.490	.076 .071
10	.850 .830	.411	.572	.636	.089 .084
12	1.055 1.035	.581	.760	.806	.089 .084
14	1.100 1.080	.650	.822	.875	.089 .084
16	1.220 1.200	.777	.947	1.002	.089 .084
18	1.350 1.330	.864	1.072	1.108	.089 .084
22	1.600 1.580	1.133	1.322	1.358	.089 .084

FIGURE 1. Dimensions and configurations - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.000050	0.00127	.112	2.84	.490	12.45	.765	19.43	1.035	26.29
.003	0.08	.118	3.00	.494	12.55	.769	19.53	1.055	26.80
.004	0.10	.134	3.40	.500	12.70	.772	19.61	1.056	26.82
.005	0.13	.138	3.51	.510	12.95	.774	19.66	1.062	26.97
.015	0.38	.144	3.66	.515	13.08	.775	19.69	1.072	27.23
.016	0.41	.148	3.76	.518	13.16	.795	20.19	1.080	27.43
.018	0.46	.160	4.07	.530	13.46	.806	20.47	1.100	27.94
.020	0.51	.164	4.17	.535	13.59	.812	20.62	1.108	28.14
.022	0.56	.194	4.93	.556	14.12	.822	20.88	1.110	28.19
.025	0.64	.198	5.03	.558	14.17	.830	21.08	1.133	28.78
.031	0.79	.204	5.18	.562	14.27	.850	21.59	1.187	30.15
.032	0.81	.224	5.69	.572	14.53	.860	21.84	1.200	30.48
.047	1.19	.265	6.73	.581	14.76	.864	21.95	1.220	30.99
.062	1.57	.281	7.14	.627	15.93	.875	22.23	1.237	31.42
.063	1.60	.290	7.37	.636	16.15	.881	22.38	1.252	31.80
.071	1.80	.329	8.36	.650	16.51	.896	22.76	1.257	31.93
.076	1.93	.340	8.64	.685	17.40	.901	22.89	1.306	33.17
.078	1.98	.351	9.92	.696	17.68	.904	22.96	1.358	34.49
.079	2.01	.388	9.86	.700	17.78	.931	23.65	1.312	33.32
.084	2.13	.394	10.01	.703	17.86	.936	23.77	1.322	33.58
.089	2.26	.400	10.16	.705	17.91	.937	23.80	1.330	33.78
.090	2.29	.411	10.44	.723	18.36	.947	24.05	1.350	34.29
.092	2.34	.413	10.49	.744	18.90	.987	25.07	1.358	34.49
.100	2.54	.418	10.62	.750	19.05	1.002	25.45	1.437	36.50
.106	2.69	.424	10.77	.754	19.15	1.007	25.58	1.580	40.13
.109	2.77	.433	11.00	.760	19.30	1.022	25.96	1.600	40.64

NOTES: (Notes 4 through 9 applicable to contacts only).

1. Dimensions are inches.
2. Metric equivlaents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are .005 (0.13 mm) for three place decimals and .01 (0.25 mm) for two place decimals. Angles $\pm$1°.
4. Contact Finish: Gold plate .000050 inch min. In accordance with MIL-G-45204. A suitable underplating shall be used. Silver shall not be used. Finish is applied after sealing.
5. Dimensions apply after plating.
6. 63 all over per ANSI B46. 1-62 unless otherwise specified.
7. Remove all burrs and sharp edges .003 to .005 R.
8. Diameters to be concentric within .004 TIR.
9. Diameters in the same cross-sectional plane to be concentric with each other .003 TIR.
10. Coupling indicator line shall be color coded red.
11. Shell size 8 will not mate with MIL-C-26500 connectors.
12. See table I for contact cavities requiring this dimension.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

This connector mates with MIL-C-83723/86, 91, 95, and 97.

C Insert arrangements and polarizing positions: See MIL-STD-1554 and table I.

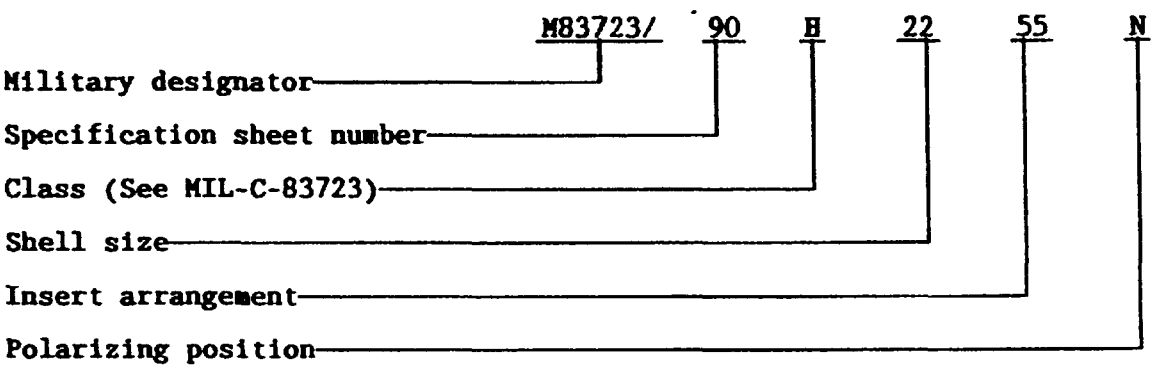
TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

Shell size	Insert arrangements	Contact cavities
8	C 03	1, 2, and 3
C 10	06	1, 2, 3, 4, 5, 6

C Connector accessories: See MIL-C-85049.

"O" ring: See MIL-C-83723.

C Part or Identifying Number (PIN) example:



CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 15, 17, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-3728-23)

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